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TIME, SPEED AND DISTANCE

TIME SPEED DISTANCE

Time Speed and Distance – Definition

Speed is a concept in motion that is all about how slow or fast an object travels. Speed is defined as the distance divided by Time. Speed, distance, and time are given to solve for one of the three variables when a piece of certain information is known.

Relationship between Time Speed and Distance

Speed = Distance/Time

Speed tells us how fast or slow an object travels and describes the distance traveled divided by the time taken to cover the distance.

Units of Time Speed and Distance

- Speed Distance and Time has different units for each of them and they are given as under
- Time: seconds(s), minutes (min), hours (hr)
- Distance: (meters (m), kilometers (km), miles, feet
- Speed: m/s, km/hr

If Distance and Time Units are known Speed Units can be derived easily.

Time Speed and Distance Conversions

- To change between km/hour to m/sec, we multiply by 5/18. So, 1 km/hour = 5/18 m/sec
- To change between m/sec to km / hour, we multiply by 18/5. So, 1m /sec = 18/5 km/hr = 3.6 km/hr

Time Speed and Distance Conversions

- Similarly, 1 km/hr = 5/8 miles/hour
- 1 yard = 3 feet
- 1 mile= 1.609 kilometer
- 1 kilometer= 1000 meters = 0.6214 mile
- 1 mile = 1760 yards
- 1 mile = 5280 feet
- 1 hour= 60 minutes= 60*60 seconds= 3600 seconds
- 1 yard = 3 feet
- 1 mph = (1 x 5280) / (1 x 3600) = 22/15 ft/sec
- 1 mph = (1 x 1760) / (1 x 3600) = 22/45 yards/sec
- For a certain distance, if the ratio of speeds is a : b, then the ratio of times taken to cover the distance is given by b : a and vice versa.

Applications of Time Speed and Distance

Check out different models of questions asked on the concept of Time Speed and Distance.

They are as under

Average Speed

Average Speed = Total Distance Travelled/Total Time Taken

Case 1: When Distance is Constant, Average Speed is given by = 2xy/x+y where x, y are two speeds at which the same distance is covered.

Case 2: When Time taken is Constant, Average Speed = (x+y)/2 where x, y are two speeds at which we traveled for the same time.

Inverse Proportionality of Speed & Time

Speed is inversely proportional to time when the distance is constant. If Speeds are in the ratio of m:n then the time taken will be n:m They are two methods of solving questions.

- Using Inverse Proportionality
- Using Constant Product Rule

Example

After traveling 60km, a train is meeting with an accident and travels at (2/3)rd of the usual Speed and reaches 30 min late. Had the accident happened 15km further on it would have reached 20 min late. Find the usual Speed?

Solution:

Using Inverse Proportionality

Here there are 2 cases

Case 1: accident happens at 60 km

Case 2: accident happens at 75 km

The difference between the two cases is only for the 15 km between 60 and 75. The time difference of 1

minutes is only due to these 15 km.

In case 1, 15 km between 60 and 75 is covered at (2/3)rd Speed.

In case 2, 15 km between 60 and 75 is covered at the usual Speed.

So the usual Time “t” taken to cover 15 km, can be found out as below. 3/2 t – t = 10 mins = > t = 20 mir

= 15 km

so Usual Speed = 15km/20min = 15km/0.33hr= 45Km/hr

Using Constant Product Rule Method

Let the actual Time taken be T

There is a (1/3)rd decrease in Speed, this will result in a (1/2)nd increase in Time taken as

Speed and Time are inversely proportional

The delay due to this decrease is 10 minutes

Thus 1/2 T= 10 and T=20 minutes

Also, Distance = 15 km

Thus Speed = 15/20 minutes = 15km/0.33hr = 45km/hr

Meeting Point

- If two people travel from points A and B towards each other they meet at point P. Distance covered by them on the meeting is AB. Time taken by both to meet is the same. Since Time is constant Distance AP and BP will be in the ratio of their Speeds.
- Consider the distance between A and B is d.
- If two people walking towards each other from A and B. when they meet for the first time they cover a distance of “d ” and when they meet for the second time they cover a distance of “3d” and when they meet for the third time they cover a distance of “5d” …..

Question: 01

A dog takes 4 leaps for every 5 leaps of a hare but 3 leaps of a dog are equal to 4 leaps of the hare. Compare their speed?

A. 12 : 16

B. 16 : 15

C. 16 : 18

D. 14 : 16

Answer:B

Explanation: 01

Let the distance covered in 1 leap of the dog be x and that covered in 1 leap of the hare by y

Then $3x = 4y$ which implies $x = 4y/3$

Ratio of speeds of dog and hare

Ratio of distances covered by them in the same time

$4x:5y = 16y/3 : 5y = 16/3 : 5$

Hence the ratio becomes 16 : 15

Question: 02

A trip to a destination is made in the following way 900 km by train at an average speed of 60 km/hr, 3000 km by pln at an average speed of 500 km/hr, 400 km by boat at an average speed of 25 km/hr, 15 km by taxi at an average speed of 45 km/hr. What is the average speed for the entire journey?

A. 100 (32/67) km/hr
B. 60 (34/89) km/hr
C. 115 (65/112) km/hr
D. 116 (72/98) km/hr

Answer:C

Explanation: 02

Total distance travelled = 900+3000+400+15 = 4315 km

Total time taken = (900/60 + 3000/500 + 400/25 + 15/45) hr = (15 + 6+ 16+ 1/ 3) hr

= 37 (1 /3)hr = 112/3 hr

Average speed for the whole journey = (4315 *3 /112) km/hr = 115(65/112) km/hr

Question: 03

One third of a certain journey was covered at the speed of 20 km/hr, one fourth at 30 km/hr and the rest at the speed of 50 km/hr. Find the average speed per hour for the whole journey.

A. 10 km/hr
B. 20 km/hr
C. 30 km/hr
D. 40 km/hr

Answer:C

Explanation: 03

Let the total distance be x km then
Distance covered at 20 km/hr = x/3 km; distance covered at 30 km/hr = x/4 km
Distance covered at 50 km/hr = [x - (x/3+x/4)] km = 5x/12 km
Total time taken = [(x/3) /20 + (x/4)/30 + (5x+12)/50] hrs
= (x/60 + x/120 + x/120) hrs = 4x/120 hrs = x/30 hrs
Average speed for the whole journey = (x* 30/x) km/hr = 30 km/hr

Question: 04

A man travelled from the village to the post office at the ate of 25 km/hr and walked back at the rate of 4 km/hr. If the whole journey took 5 hrs 48 mins, find the distance of the post office from the village?

A. 5 km/hr

B. 10 km/hr

C. 15 km/hr

D. 20 km/hr

Answer:D

Explanation: 04

Average speed = (2xy / x+y) km/hr = (2*25*4 / 25+4) km/hr = 200/29 km/hr
Distance travelled in 5 hrs 48 mins, i.e $5\frac{4}{5}$ = (200/29 * 29/5) km = 40 km
Distance of the post-office from th village = (40/2) = 20 km

Question: 05

A person reaches his destination 40 mins late if his speed is 3 km/hr, and reaches 30 mins before time if his speed is 4 km/hr. Find the distance of his destination from his starting point?

A. 14 km

B. 15 km

C. 16 km

D. 17 km

Answer:A

Explanation: 05

Let the required distance be x km

Difference in the times taken at two speeds = 70 min = 7/6 hr

$x/3 - x/4 = 7/6$

$4x - 3x = 14$

$x = 14$ km

Hence therequired distance is 14km

Question: 06

A carriage driving in a fog passed a man who was walking at the rate of 3 km/hr in the same direction. He could see the carriage for 4 mins and it was visible to him uto a distance of 100m. What was the speed of the carriage?

A. 3/ 8 km/hr

B. 9 /2 km/hr

C. 6/ 8 km/hr

D. 9/13 km/hr

Answer:B

Explanation: 06

Let the speed of the carriage be x km/hr. Then relative speed = (x- 3) km/hr
Distance covered in 4 min, i.e 1/15 hr at relative speed = 100m = 1/10 km
(x-3) = 1/10 * 15 = 3/ 2
x= 3 + 3/2 = 9/2
Hence the speed of the carriage = 9/2 km/hr

Question: 07

Two boys A and B start at the same time to ride from Delhi to Meerut, 60 km away. A travels 4 km an hour slower than B.B reaches Meerut and at once turns back meeting A 12 km from Meerut. Find A,s speed?

A. 5 km/hr

B. 8 km/hr

C. 9 km/hr

D. 10 km/hr

Answer:B

Explanation: 07

Let A’s speed = x km/hr, then B’s speed = x+4 km/hr
Clearly time taken by B to cover (60 +12) = 72 km= time taken by A to cover (60 -12)
i.e 48 km
72/x+4 = 48/x
72x = 48x + 192
24x = 192
x= 8
Hence A’s speed = 8 km/hr

Question: 08

A thief is spotted by a policeman from a distance of 100m. When the policeman starts the chase, the thief also starts running. If the speed of the thief be 8 km/hr and that of the policeman 10 km/hr how far the theif will have run before he is overtaken?

A. 200 m

B. 300 m

C. 400 m

D. 500 m

Answer:C

Explanation: 08

Relative speed of the policeman = (10 -8) = 2 km/hr
Time taken by policeman to cover 100m = (100/ 1000 *2) = 1/20 hr
In 1/20 hr the theif covers a distance of (8/20) km = 2/ 5 km = 400 m

Question: 09

A man takes 6 hrs 30 min in going by a cycle and coming back by scooter. He would have lost 2 hours 10 min by going on cycle both ways. How long would it take him to go by scooter both ways?

A. 3 hr 20 min

B. 4 hr 20 min

C. 5 hr 25 min

D. 6 hr 50 min

Answer:B

Explanation: 09

Let the distance be x km, then
Time taken to cover x km by cycle + time taken to cover x km by scooter = 6 hr 30 min
Time taken to cover 2x km by cycle + time taken to cover 2x km by scooter = 13 hr
But time taken to cover 2x km by cycle = 8 hr 40 min
Time taken to cover 2x km by scooter = 13 hr- 8 hr 40 min = 4 hr 20 min

Question: 10

Two places A and B are 80 km apart from each other on a highway. A car starts from A and another from B at the same time. If they move in the same direction they meet each other in 8 hours. If they move in opposite directions towards each other, they meet in 1 hour 20 mins. Determine the speeds of the cars?

A. 20, 30 km/hr
B. 30, 45 km/hr
C. 35, 25 km/hr
D. 40 ,50 km/hr

Answer:C

Explanation: 10

Let their speeds be x km/hr and y km/hr respectively
Then 80/x-y = 8 => x-y =10 …….(1)
And 80/x+y = 1 (1/ 3)= 4/3 => x+y = 60 ……..(2)
Adding 1 and 2 we get 2x= 70 x=35 km/hr
Hence y = 25 km/hr

Question: 11

A goods train leaves a station at a certain time and at a fixed speed. After 6 hours, an express train leaves the same station and moves in the same direction at a uniform speed of 90 km/hr. This train catches up the goods train in 4 hours. Find the speed of the goods train?

A. 28 km/hr

B. 30 km/hr

C. 34 km/hr

D. 36 km/hr

Answer:D

Explanation: 11

Let the speed of the goods train be x km/hr
Distance covered by goods train in 10 hours = distance covered by express train in 4 hours
$10x = 4 * 90$
$x = 36$ km/hr

Question: 12

The average speed of a bus in one-third of the speed of a train. The train covers 1125 km in 15 hours. How much distance will the bus cover in 36 minutes?

A. 12 km

B. 18 km

C. 21 km

D. None of these

Answer:D

Explanation: 12

Speed of the train = 1125 /15 km/hr -= 75 km/hr
Speed of the bus = 1/ 3 *75 km/hr = 25 km/hr
Distance covered by the bus in 60 min = 25 km
Distance covered by the bus in 36 min = 25 *36 / 60 = 15 km

Question: 13

Jane travelled 4/7 as many miles on foots as by water and 2/ 5 as many miles on horseback as by water. If she covered a total of 3036 miles, how many miles did she travel on foot?

A. 1540

B. 880

C. 756

D. 616

Answer:B

Explanation: 13

Suppose jane travelled x miles by water, (4x/7) miles on foot and (2x/5) miles on horseback
Then, $x + 4x/7 + 2x/5 = 3036$
$69x/35 = 3036$
$x = 3036 * 35 / 69 = 1540$
Distance travelled on foot = 4/7 *1540 miles = 880 miles

Question: 14

Deeps rides her bike at an average speed of 30 km/hr and reaches her destination in 6 hours. Hema covers the same distance in 4 hours. If deepa increases her average speed by 10 km/hr and Hema increases her average speed by 5 km/hr what would be the difference in their time taken to reach the destination?

A. 40 mins

B. 45 mins

C. 54 mins

D. 1 hour

Answer:C

Explanation: 14

Deepa"s original speed = 30 km/hr
Deepa"s new speed = 30 + 10 = 40 km/hr
Distance = 30 *6 = 180 km
Hema"s original speed= 180 /4 km/hr= 45 km/hr
Hema's new speed = 45 +5 = 50 km/hr
Difference in time = (180/40 - 180/50) =9/10 hrs= 9/10 *60 = 54 min

Question: 15

A monkey climbing up a pole ascends 6 meters and slips 3 metres in alternate mins. If the pole is 60 m high, how long will it take the money to reach the top?

A. 31 min

B. 33 min

C. 35 min

D. 37 min

Answer:D

Explanation: 15

Net height ascended in 2 min = 6 -3 = 3 m
Net height ascended in 36 min = (3/2 *36) = 54 m
In the 37th min, the monkey ascends 6m and reaches the top
Hence the total time taken = 37 mins

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THANK YOU

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